OurMocks

2027 CYCLE

MOCK · PRACTICE · CRACK

CUET UG

Biology

Principles of Inheritance and Variation

Official Syllabus Unit VII · Class 12 · CUET UG 2027 Cycle Notes

INSIDE
Concept Notes

INSIDE
Solved Examples

INSIDE
Practice MCQs + Solutions

Aligned to the latest official syllabus · Concept-first explanations

www.ourmocks.com

Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) Biology syllabus 2026, the current 2027 baseline, Unit VII)	Skills the questions test
Mendel's law of inheritance	Inheritance of one gene — law of dominance, law of segregation (test and back cross).	Setting up P $\rightarrow$ F_1 $\rightarrow$ F_2 crosses, walking Punnett squares, quoting the 3 : 1 and 1 : 2 : 1 ratios.
Law of dominance	Inheritance of one gene — law of dominance.	Spotting which allele dominates which in the F_1, and where the hidden trait reappears.
Law of segregation — test and back cross	Inheritance of one gene — law of segregation (test and back cross).	Using a 1 : 1 test cross to prove heterozygosity; telling a test cross from a back cross.
Deviation from Mendelism	Deviation from Mendelism — incomplete dominance and co-dominance.	Telling a blend from a both-show case; snapdragon pink and AB blood group examples.
Multiple alleles and blood groups	Multiple alleles and inheritance of blood groups.	Writing the six ABO genotypes, running blood-group crosses, ruling out impossible children.
Inheritance of two genes	Inheritance of two genes — law of independent assortment, dihybrid cross.	Building the 4 $\times$ 4 grid, quoting 9 : 3 : 3 : 1 and the 1 : 1 : 1 : 1 dihybrid test cross.
Chromosomal theory of inheritance	Chromosomal theory of inheritance.	Matching chromosome behaviour in meiosis to Mendel's factor pairs.
Linkage and recombination	Linkage and recombination.	Predicting which genes travel together; counting parental vs recombinant types.
Polygenic inheritance	Polygenic inheritance.	Explaining continuous variation, bell-shaped F_2 ranges and gene-plus-environment effects.

OurMocks

Sub-topic	Official syllabus pointer (NTA CUET(UG) Biology syllabus 2026, the current 2027 baseline, Unit VII)	Skills the questions test
Sex determination and disorders	Sex determination in humans, birds and honey bee; pedigree analysis, Mendelian disorders (colour blindness, haemophilia, sickle-cell anaemia, phenylketonuria and thalassemia), chromosomal disorders in humans (aneuploidy, Down's syndrome, Turner's syndrome and Klinefelter's syndrome).	Matching karyotypes to syndromes, reading family trees, X-linked reasoning.

EXAM PATTERN

CUET UG pattern (NTA official information bulletin and syllabus, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The Biology paper maps strictly to NCERT Class 12, and Unit VII starts here: Mendel's ratios are the grammar of genetics — most questions in this unit read them, bend them, or trace them to chromosomes.

Mendel and his Peas

Every family carries a running debate — whose nose is this, why did grandfather's dimples skip a generation. Gregor Mendel turned that debate into a counting problem, and the answers still run the whole of genetics. He grew peas in a monastery garden, counted tens of thousands of plants, and wrote down ratios that examiners quote to this day.

- **Who he was:** Gregor Johann Mendel, a monk at the monastery in Brno (then in Austria-Hungary, now in Czechia), taught science and gardened.
- **When he worked:** the pea experiments ran from 1856 to 1863; he presented the results in 1865 and published them in 1866.
- **Forgotten, then found:** the paper gathered dust until 1900, when three botanists — de Vries, Correns and von Tschermak — rediscovered it independently.
- **His big idea:** traits are carried by discrete **factors** (what we now call genes) that pass unchanged from parent to offspring.

His method in five steps

1. **Choose pure lines:** keep one variety self-pollinating until it reliably produces only tall plants, another only dwarf ones.
2. **Cross two contrasting pure plants:** brush pollen from the dwarf onto the tall — this is the P (parental) generation.
3. **Grow the F_1 and look:** record which form every offspring shows.
4. **Let the F_1 self-pollinate:** raise the F_2 and **count** both forms.

5. **Do the arithmetic:** express the counts as a ratio, and repeat with large numbers so chance cannot distort the answer.

Why the garden pea was the right choice

Mendel's success was not luck — the material was chosen almost perfectly. Pea plants give clean, countable answers where most other plants give mush.

Advantage of peas	What it meant for Mendel	Why it matters for a clean ratio
Seven clear contrasting traits	Seed shape (round or wrinkled), seed colour (yellow or green), flower colour (violet or white), pod shape (inflated or constricted), pod colour (green or yellow), flower position (axial or terminal), plant height (tall or dwarf).	No halfway forms — every plant lands in one bin, so counts are sharp.
True-breeding lines	Each variety, self-pollinated for many generations, gave only its own form of the trait.	The parents were pure (homozygous) — the starting point was known exactly.
Natural self-pollination	Pea flowers stay closed, so unwanted pollen rarely sneaks in; he could also open flowers and hand-cross.	He controlled which plant fathered which seed — no surprise fathers.
Many offspring, short cycle	Hundreds of seeds per plant, and a new generation every season.	Big numbers make ratios like 2.96 : 1 settle visibly at 3 : 1.

The vocabulary, in plain words

- **Gene:** the unit of inheritance — Mendel's "factor" — a stretch of DNA that decides one character.
- **Allele:** one version of a gene. The height gene has a **tall** allele and a **dwarf** allele.
- **Locus:** the gene's fixed address on a chromosome; both alleles of a pair sit at the same address on the two homologous chromosomes.
- **Genotype:** the letter pair written inside the organism, e.g. Tt.
- **Phenotype:** what you actually see, e.g. tall.
- **Homozygous and heterozygous:** two identical alleles (TT or tt) versus two different ones (Tt).
- **Dominant and recessive:** the allele that shows in the heterozygote (tall, T) is dominant; the one that hides (t, dwarf) is recessive.

The monohybrid cross, step by step

A **monohybrid cross** follows one character — here, plant height. Mendel crossed pure tall (TT) with pure dwarf (tt) plants and watched three generations.

- **Parents (P generation):** TT tall × tt dwarf. Each parent is pure, so every gamete from the tall parent carries T, every gamete from the dwarf parent carries t.
- **F_1 generation:** all offspring are Tt — and **all are tall**. The dwarf look has vanished, but the t allele is still riding inside every plant.
- **F_1 gametes:** each Tt plant makes two kinds of gametes, T and t, in equal numbers.
- **Self-pollinate the F_1:** a **Punnett square** — a grid listing every female gamete against every male gamete — collects all sixteen… four pairings, as shown below.

F_1 gametes	T	t
T	TT — tall	Tt — tall
t	Tt — tall	tt — dwarf

- **F_2 genotype ratio:** $1TT : 2Tt : 1tt$ — the classic 1 : 2 : 1.
- **F_2 phenotype ratio:** 3 tall : 1 dwarf — three genotypes look tall, one looks dwarf.
- **The surprise Mendel counted:** in 14,349 seeds of the seed-shape cross, he got 5,474 round and 1,850 wrinkled — a ratio of 2.96 : 1. Big numbers, clean ratio.

EVERYDAY EXAMPLE — TWO COIN FLIPS

Flip two coins and record heads (H) or tails (T). The possible pairs are HH, HT, TH and TT — exactly one pure pair, two mixed pairs, one opposite pure pair. Now map H to the tall allele and T to the dwarf allele: the coin table is the Punnett square, and the 1 : 2 : 1 falls out of arithmetic, not magic. The mapping is honest: each parent gives one random coin (gamete), and the pairing happens at fertilisation.

The Three Laws

Mendel never wrote the word “gene”, yet three short statements from his 1866 paper are now taught as his three laws. Learn each law as one line plus one cross that proves it — that is exactly how questions test them.

Law of dominance

- **The statement:** in a contrasting pair of factors, one factor **dominates** — it alone shows in the F_1 generation.
- **The proof in the cross:** TT × tt gave an all-tall F_1 the dwarf factor did not appear, but was not destroyed.
- **The comeback:** self-pollinating that F_1 let the dwarf trait reappear in the F_2 — hidden traits survive, they only hide.
- **Watch the wording:** dominance is a relationship **between two alleles**, not a property one allele carries around on its own.

Law of segregation

- **The statement:** the two members of a factor pair separate (**segregate**) when gametes form; every gamete receives exactly one.

- **Why gametes are "pure":** a Tt plant makes only T or t gametes — never a blended "Tt" gamete. This law is also called the **law of purity of gametes**.
- **No blending, ever:** the alleles stay distinct through every generation; they only meet again at fertilisation.
- **The 50 : 50 split:** a heterozygote makes T and t gametes in equal numbers — that is why ratios in the next generation come out whole.

Law of independent assortment

- **The statement:** when two pairs of traits are followed together, each pair segregates **independently** — one pair does not disturb the other.
- **The proof in the cross:** the dihybrid cross (Section 5) gives four gamete types in equal numbers and a 9 : 3 : 3 : 1 F_2 — the signature of free assortment.
- **The fine print:** independence belongs to genes on **different chromosomes** (or very far apart on the same one). Section 6 shows what happens when genes sit close together — linkage bends this law.

Test cross and back cross

Point of difference	Test cross	Back cross
Definition	F_1 offspring crossed with its **recessive** parent or an individual genetically similar to it.	F_1 offspring crossed back with **its own parent** or an individual genetically similar to its parent.
Partner chosen	Always the homozygous recessive (tt) — the only partner that cannot mask anything.	The parent (or a parent-like plant) — dominant or recessive, whatever the parent was.
Typical result	1 : 1 ratio of the two phenotypes when the F_1 is heterozygous.	No fixed ratio — it depends which parent is used; often the aim is to keep the parent's traits.
What it tells you	An unknown dominant plant's genotype: 1 : 1 result means heterozygous, all-tall result means homozygous.	Used by breeders to attach a useful trait (like high yield) back onto a parent-like background.

WORKED EXAMPLE — THE 1 : 1 TEST CROSS

A tall pea plant of unknown genotype is crossed with a dwarf plant (tt). Half the offspring are tall, half are dwarf. What was the tall parent?

- **Read the result:** dwarf offspring are tt, so each parent contributed a t.
- **Follow the tall offspring:** they must be Tt — a T from the tall parent and a t from the dwarf parent.
- **So the tall parent made two gamete types:** T and t — its genotype was Tt (heterozygous).

Cross on paper: $Tt \times tt \rightarrow$ gametes T, $t \times t$, $t \rightarrow$ offspring $1Tt$ (tall) : $1tt$ (dwarf). That is 50% tall and 50% dwarf — a 1 : 1 result **proves** the tall parent was heterozygous. A homozygous TT parent would have given only tall offspring.

One-minute judgements

Decide each statement before reading the check line — these are the exact sentences MCQ options are built from.

Statement	Verdict	Check line
A Tt plant makes some "Tt" gametes.	False	Segregation: gametes carry **one** allele — purity of gametes.
After $TT \times tt$, the dwarf trait is destroyed forever.	False	It only hides; the F_2 shows dwarf plants again.
Mendel's factors blend inside the hybrid.	False	Factors stay distinct — pure forms reappear in the F_2.
A test cross pairs the unknown plant with a dominant partner.	False	The partner must be **recessive**, so it masks nothing.

CROSS TOOLKIT — RATIOS YOU WILL QUOTE

Monohybrid F_2 (Tt × Tt)

3 : 1 phenotypes

1 : 2 : 1 genotypes

Monohybrid test cross (Tt × tt)

1 : 1

Dihybrid F_2 (RrYy × RrYy)

9 : 3 : 3 : 1

Dihybrid test cross (RrYy × rryy)

1 : 1 : 1 : 1

When Mendel Bends: Incomplete and Co-dominance

Not every cross reads 3 : 1. Grow a red snapdragon next to a white one, cross them, and the F_1 flowers come out **pink** — no dominance, no hiding, just a middle shade. Mendel's laws describe what happens when one allele fully covers the other; this section shows what happens when the mask slips. Both examples here are in the official syllabus pointer, and both are classic one-mark checks.

Incomplete dominance: the blend

- **Definition:** in **incomplete dominance** (also called partial dominance), the F_1 shows a phenotype **between** the two parents — neither allele fully covers the other.
- **The textbook case: snapdragon** (**Antirrhinum**, the dog flower). A pure red flower (RR) crossed with a pure white one (rr) gives an F_1 of all pink (Rr) flowers.
- **Self-pollinate the pink F_1:** the F_2 splits into 1 red : 2 pink : 1 white.
- **Why the ratio changes shape:** the genotype ratio is still the old 1 : 2 : 1 — but now each genotype wears its own visible coat, so the **phenotypic ratio becomes 1 : 2 : 1 too**. The heterozygote no longer looks like the dominant homozygote.
- **Same story in rice and peas' cousins:** four o'clock plant (**Mirabilis**) shows the identical red-pink-white pattern.

F_1 gametes	R	r
R	RR — red	Rr — pink
r	Rr — pink	rr — white

Co-dominance: both speak at once

- **Definition:** in **co-dominance**, both alleles of the pair express themselves **simultaneously** in the F_1 — nothing blends, nothing hides.
- **The textbook case:** the AB blood group. A person carrying I^A and I^B makes **both** surface sugars on red blood cells, so the phenotype is AB.
- **The visible pattern:** in a cross, both parental characters appear **together** in the offspring — like a striped flag, not a mixed paint bucket.
- **Same law underneath:** segregation still holds. The F_2 ratios just look different because the heterozygote has its own unmistakable look.

The three dominance patterns side by side

Feature	Complete dominance	Incomplete dominance	Co-dominance
F_1 looks like	The dominant parent — recessive allele fully hidden.	Halfway between the two parents — a new intermediate.	Both parents' characters at once, side by side.
Classic example	Tall × dwarf pea → all tall F_1.	Red × white snapdragon → pink F_1.	$I^A \times I^B$ parents → AB blood group child.
F_2 phenotypic ratio	3 : 1	1 : 2 : 1	1 : 2 : 1 (each phenotype matches one genotype class).
What the heterozygote does	Copies the dominant homozygote exactly.	Shows its own intermediate phenotype.	Shows both alleles' products together.

Feature	Complete dominance	Incomplete dominance	Co-dominance
Mendel's law affected	None — the standard case.	Dominance bends; segregation stands.	Dominance bends; segregation stands.

EVERYDAY EXAMPLE — PAINT BUCKET VS FLAG

Mix red paint and white paint and you get pink — that is **incomplete dominance**: the mixture is a genuinely new, intermediate shade, and you cannot un-mix it by looking. Fly a red flag next to a white flag on the same pole and you see **both colours**, each in its own stripe — that is **co-dominance**: both alleles express, neither averages out. The mapping is honest: paint = blended expression level, flag = two products displayed together. Blood group AB is two flags on one cell.

WORKED EXAMPLE — READING A PINK-FLOWER F#SUB[2]

Two pink snapdragons are crossed. Predict the F_2.

- **Parents:** Rr (pink) × Rr (pink).
- **Gametes:** each plant makes R and r in equal numbers.
- **F_2:** $1RR$ (red) : $2Rr$ (pink) : $1rr$ (white).

So you expect 50% pink, 25% red and 25% white among the offspring. If a question says "out of 40 F_2 plants", the expected split is 10 red : 20 pink : 10 white — divide the 40 into 4 parts of 10, then give each phenotype its share. One more check: if every F_1 plant of a four o'clock cross was pink, the parents must have been pure red and pure white — two pinks could not have produced all-pink offspring.

Which pattern is it? Three quick cases

- **Case 1 — F_1 all tall, F_2 in 3 : 1:** complete dominance; the recessive allele was hidden, not changed.
- **Case 2 — F_1 all pink, F_2 in 1 red : 2 pink : 1 white:** incomplete dominance; the heterozygote shows its own intermediate shade.
- **Case 3 — every F_1 shows both parental characters at once:** co-dominance; both alleles produce visible products together.

Why incomplete dominance is not "blending inheritance"

- **The factors stay separate:** pink plants still carry R and r as individual alleles — nothing merged inside them.
- **The proof:** self-pollinating the pink F_1 recovers pure red (RR) and pure white (rr) in the F_2. True blending could never give back the two extremes.
- **The exam point:** "blending" language describes the **phenotype** only — the genotype remains fully particulate.

Multiple Alleles and Blood Groups

Your blood-group card is a genetics lesson printed in one line. Blood group comes from **one gene** that has **more than two allelic forms** in the human population — a situation called **multiple alleles**. Mendel saw only two versions of each trait; the ABO system has three. The rules of inheritance stay exactly the same — meiosis still hands out one allele per gamete.

One gene, many forms

- **Definition: multiple alleles** means a gene exists in more than two allelic forms in a population — but **any one individual still carries only two**, one on each homologous chromosome.
- **The ABO gene:** the I gene has three alleles — I^A, I^B and i.
- **What each allele does:** I^A and I^B each direct a different sugar layer on the red blood cell surface; i builds no sugar at all.
- **Dominance pattern:** I^A and I^B each dominate i, but between themselves they are **co-dominant** — both express in an AB person.
- **Why more than two alleles matters:** it multiplies the possible genotypes — three alleles give six genotype combinations.

Six genotypes, four phenotypes

Phenotype (blood group)	Possible genotypes	Genotype logic in one line
A	I^AI^A or I^Ai	Two I^A, or one I^A with a silent i.
B	I^BI^B or I^Bi	Two I^B, or one I^B with a silent i.
AB	I^AI^B only	Co-dominance — both sugars displayed; the only two-allele phenotype.
O	ii only	Both copies are the silent i — no surface sugar at all.

- **Count check:** I^AI^A, I^Ai, I^BI^B, I^Bi, I^AI^B, ii — six genotypes, four phenotypes. Questions love asking you to reproduce this list.
- **O is not a "missing" gene:** i is a real, working allele that simply makes no detectable sugar; it is recessive because it hides behind I^A or I^B.

A blood-group cross, worked in full

WORKED EXAMPLE — GROUP A PARENT × GROUP B PARENT

A father is group A with genotype I^Ai, and the mother is group B with genotype I^Bi. Both parents are heterozygous. What can the children be?

- **Father's gametes:** I^A and i — one per sperm.

- **Mother's gametes:** I^B and i — one per egg.

Mother's gametes	I^A	i
I^B	$I^A I^B$ — AB	$I^B i$ — B
i	$I^A i$ — A	ii — O

- **Result:** AB, B, A and O children are all possible — in a 1 : 1 : 1 : 1 ratio among the zygotes.
- **The striking part:** two "non-O" parents can have an O child, because both carried the silent i all along.

Reverse check — the classic trap: an AB parent ($I^A I^B$) × an O parent (ii) gives only $I^A i$ (A) and $I^B i$ (B) children. The AB parent always passes either I^A or I^B, so an O child is impossible — there is no i to collect from that side.

The Rh factor and erythroblastosis foetalis

- **Rh factor:** a second blood antigen system; the **Rh$^+$** condition (antigen D present) is dominant over **Rh$^-$** (antigen absent).
- **The risky pregnancy:** an Rh$^-$ mother carrying an Rh$^+$ baby can be sensitised in the first pregnancy when foetal blood mixes with hers.
- **The second pregnancy:** her memory immune system now makes **anti-D antibodies** that cross the placenta and destroy the foetus's red blood cells.
- **The name:** this newborn condition is **erythroblastosis foetalis** — anaemia and jaundice caused by the mother's antibodies.
- **The fix:** anti-D injections given to the Rh$^-$ mother prevent sensitisation.
- **Donor shorthand:** group O is the **universal donor** (no A, B sugars to trigger), group AB the **universal recipient** (no anti-A or anti-B antibodies to react).

Who can give blood to whom

The ABO and Rh systems together produce eight common blood types. The matching rule in one line: **give no antigen the recipient's antibodies can attack.**

Blood type	Antigens on the red cells	Can give red cells to
O$^-$	None — no A, B or Rh antigen	Everyone — the safest universal red-cell donor
A	A antigen	A and AB recipients
B	B antigen	B and AB recipients
AB$^+$	A, B and Rh antigens	Only AB$^+$ recipients — but can receive from anyone

- **Why O works everywhere:** an O red cell wears no A or B sugar, so no recipient's anti-A or anti-B antibody finds anything to attack.

- **Why AB accepts everything:** an AB person makes neither anti-A nor anti-B antibody, so no donor sugar is foreign to them.
- **Rh joins the rule:** Rh^- recipients must not receive Rh^+ red cells, because they can learn to make anti-D.

Two Genes Together: The Dihybrid Cross

So far we have tracked one character at a time. Real organisms carry thousands of genes, so Mendel asked the obvious next question: what happens to two characters at once? He crossed pure round-yellow peas with pure wrinkled-green peas and counted the grandchildren. The answer — 9 : 3 : 3 : 1 — is the single most quoted ratio in genetics.

Setting up the cross

- **Parental cross:** pure round-yellow ($RRYY$) × pure wrinkled-green ($rryy$). Round dominates wrinkled; yellow dominates green.
- **F_1:** all offspring are $RrYy$ — round and yellow, exactly like the dominant parent in looks.
- **The key question:** does the F_1 make only the parental gamete types (RY and ry), or all combinations?
- **Mendel's observation:** the F_1 makes **four** gamete types — RY, Ry, rY, ry — in equal numbers. Seed shape and seed colour are chosen independently, like two separate coin flips.

The 4 × 4 Punnett grid

Self-pollinating the F_1 means 4 female gamete types × 4 male gamete types

16 equally likely boxes. The figure below shades the 16 boxes into four

phenotype groups — count the boxes and the ratio reads straight off the colours.

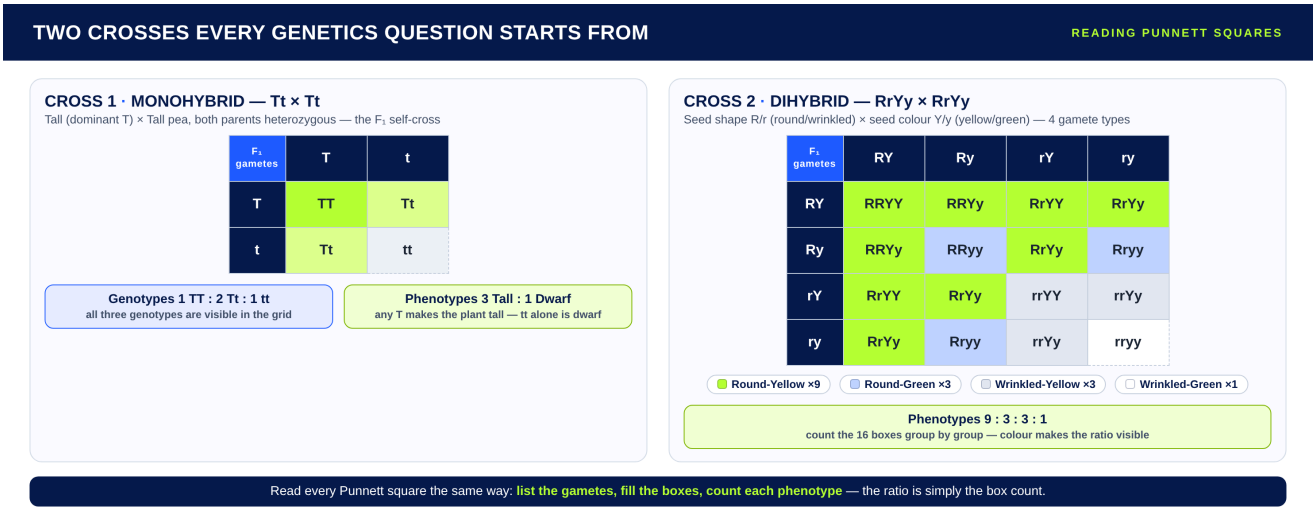


Figure 1: The two cross-grids every genetics question starts from — monohybrid Tt × Tt and dihybrid RrYy × RrYy, with their ratios visible as shaded box counts.

- **Round-yellow** ($R_Y_$): 9 boxes — every genotype with at least one R and one Y.

- **Round-green** (R_yy): 3 boxes.
- **Wrinkled-yellow** ($rrY_$): 3 boxes.
- **Wrinkled-green** ($rryy$): 1 box — the double recessive.
- **The F_2 phenotypic ratio:** 9 : 3 : 3 : 1. Two of the groups (round-green and wrinkled-yellow) are **new combinations** that neither parent showed.

Genotype	Box count	Genotype	Box count
$RRYY$	1	$RRyy$	1
$RRYy$	2	$Rryy$	2
$RrYY$	2	$rrYY$	1
$RrYy$	4	$rrYy$	2
		$rryy$	1

- **Nine genotype classes in all:** the four subscript patterns give 9 genotypes whose box counts sum to 16. Sum check: $1 + 2 + 2 + 4 + 1 + 2 + 1 + 2 + 1 = 16$.
- **Independent assortment in plain words:** picking "round or wrinkled" does not disturb the picking of "yellow or green". Each gene pair sorts into gametes as if the other pair did not exist.

Drawing the 4 × 4 grid fast in an exam

1. **Write the four gamete types once across the top and once down the side:** RY, Ry, rY, ry — always in that order, so no box is missed.
2. **Fill each box with one letter from each pair:** take one shape letter and one colour letter per parent, e.g. $rY \times Ry$ gives $rR\ Yy$ — write it as $RrYy$ (capital first).
3. **Circle groups by phenotype and count:** 9 of one kind, then 3, 3 and 1.

- **General rule worth carrying:** a plant heterozygous for n independent genes makes 2^n gamete types — 2 genes give 4, three genes give 8, and the grid grows as $2^n \times 2^n$.

The dihybrid test cross

WORKED EXAMPLE — RRYY × RRYY GIVES 1 : 1 : 1 : 1

Cross the dihybrid F_1 ($RrYy$) with the double recessive ($rryy$) — the dihybrid **test cross**.

- **F_1 gametes:** RY, Ry, rY, ry — four types, equal in number.
- **Recessive parent's gametes:** only ry — it cannot mask anything.
- **Offspring:** $RrYy$ (round-yellow), $Rryy$ (round-green), $rrYy$ (wrinkled-yellow), $rryy$ (wrinkled-green).

Result: 1 : 1 : 1 : 1 — four phenotypes in equal counts. This ratio is the experimental proof that the F_1 really made four gamete types in equal numbers, which is the law of independent assortment made visible. If the counts came out lopsided (mostly round-yellow and wrinkled-green), you would conclude the two genes were **linked** — the plot of Section 6.

MENDEL'S THREE LAWS — ONE LINE EACH

Law of dominance: in a contrasting pair, one factor dominates the other in the F_1.

Law of segregation: paired factors separate at gamete formation; each gamete gets one — never both.

Law of independent assortment: one factor pair sorts into gametes independently of another pair.

Chromosomes Carry Genes: Linkage and Recombination

Mendel died in 1884 never knowing where his factors lived. In 1902 two scientists working with microscopes found the address — the chromosomes — and genetics joined hands with cell biology. This section explains why some genes obey the law of independent assortment strictly, and why others quietly ignore it.

The chromosomal theory of inheritance

- **Who proposed it:** Walter Sutton and Theodor Boveri, working independently around 1902, united the behaviour of chromosomes with Mendel's factors.
- **The core claim:** chromosomes are the **physical carriers** of Mendel's factors — the genes ride on chromosomes, in fixed positions.
- **The parallel that convinced everyone:** chromosome behaviour in meiosis matches factor behaviour in crosses, point for point.

Mendel's factors	Chromosomes in meiosis
Factors come in pairs — one from each parent.	Chromosomes come in homologous pairs — one from each parent.
Factor pairs separate; each gamete gets one factor of a pair.	Homologous pairs separate in meiosis I; each gamete gets one chromosome of a pair.
Different pairs assort independently.	Different chromosome pairs line up and separate independently of each other.
Fertilisation restores the paired condition.	Egg + sperm restore the diploid set (46 in humans).

Linkage: genes that travel together

- **Definition: linkage** is the physical association of two or more genes on the same chromosome — they inherit as a group.
- **The consequence:** linked genes do **not** assort independently; they tend to pass into the same gamete, so parental combinations dominate the offspring.
- **Tightness matters:** the closer two genes sit, the less often they separate — **tightly linked** genes give very few recombinant offspring.

- **Loose linkage:** genes far apart on the same chromosome separate more often, so more recombinants appear.

Recombination: crossing over breaks the team

- **Definition: recombination** is the generation of new gene combinations — the result of **crossing over**, the exchange of chromosome segments between homologous chromosomes during prophase I of meiosis.
- **Frequency reflects distance:** the further apart two genes are, the more space there is for a crossover between them — so **recombination frequency rises with distance**. Geneticists exploit this to map gene positions.
- **The tension with Mendel:** independent assortment predicts 50 : 50 parental and recombinant gametes; tight linkage gives mostly parental. The further genes drift apart, the closer results come back to the 50 : 50 Mendelian look.

Morgan's fruit flies: the evidence

- **The worker:** Thomas Hunt Morgan and his students, at Columbia University, took genetics into the lab with the fruit fly **Drosophila melanogaster** — short life cycle, four chromosome pairs, hundreds of offspring.
- **The discovery:** a single **white-eyed** male appeared in a red-eyed culture. Crossing experiments showed white eye appeared almost only in **males** — the gene sat on the **X chromosome**, an **X-linked** (sex-linked) gene.
- **Why it mattered:** a concrete gene was pinned to a concrete chromosome — the chromosomal theory had its proof, and Morgan's group went on to build chromosome maps from recombination frequencies.
- **Linkage groups:** genes on one chromosome form a **linkage group**; Drosophila has 4 chromosome pairs and 4 linkage groups — the counts match, as the theory demands.

Tight or loose: how linkage changes the numbers

Point	Tightly linked genes	Loosely linked / unlinked genes
Distance on the chromosome	Very close together — almost no room for a crossover.	Far apart, or on different chromosomes.
Test-cross result	Mostly the two parental types; recombinants rare.	Close to 1 : 1 : 1 : 1 — recombination approaches 50%.
F_2 ratio look	Distorted away from 9 : 3 : 3 : 1, with parental pairs heavy.	Clean 9 : 3 : 3 : 1 — the law of independent assortment reads normally.
Map distance read-out	Low recombination frequency — small map distance.	High frequency (up to 50%) — large distance or separate chromosomes.

EVERYDAY EXAMPLE — BEADS ON ONE STRING

Picture each chromosome as a string of beads. Genes on **different strings** shuffle freely when the strings are dealt out — full independent assortment. Genes on the **same string** stay neighbours unless the string is cut and rejoined between them (crossing over). Tight neighbours — beads almost touching — are nearly always dealt together; beads at opposite ends are cut apart often. The mapping is honest: string = chromosome, bead = gene, cut-and-rejoin = crossing over, deal = gamete formation.

Polygenic Inheritance and Sex Determination

Look around a classroom: nobody is exactly "tall" or "short" — heights form a smooth spread. Traits like this are not governed by one gene with a dominant and a recessive allele; they are built by **many genes adding small effects**, with the environment stirring the mix further. This section covers that, plus the completely different question of how a body becomes male or female.

Polygenic inheritance: many genes, one trait

- **Definition: polygenic inheritance** is the control of one trait by three or more genes, each contributing an additive dose — plus environmental effects on top.
- **The signature: continuous variation**. Instead of two or three neat categories, you get a full range of shades — no clean ratios anywhere.
- **Human skin colour (the NCERT case):** worked out with three genes A, B, C — the dominant alleles each add pigment. A person with $AABBCC$ is very dark; $aabbcc$ is very light; between them, genotypes with 5, 4, 3, 2 or 1 dominant alleles build the full ladder of shades.

Dominant alleles of the six	Shade class	Sample genotype
6	Very dark skin	$AABBCC$ — the pigment ceiling
4	Medium-dark	$AaBBCc$ — four doses of pigment
2	Medium-light	$AabbCc$ — two doses
0	Very light	$aabbcc$ — the pigment floor

- **Wheat kernel colour (Nilsson-Ehle):** crossing red-kernel with white-kernel wheat gave an F_1 of intermediate kernels and an F_2 spread across five shades — the classic early demonstration of polygenic inheritance.
- **Why no 3 : 1:** each of the several gene pairs segregates on its own; their products pile up, and the F_2 forms a **bell-shaped** distribution with the parental extremes rare at the two tips.
- **More examples:** human height, human intelligence-related measures — anything measured on a smooth scale is polygenic at heart.

Where the bell shape comes from

WORKED EXAMPLE — COUNTING THE F#SUB[2] OF A THREE-GENE CROSS

Cross a very dark person (*AABBCC*) with a very light one (*aabbcc*), then let the trihybrid F_1 self-cross conceptually.

- **F_1:** *AaBbCc* — every gene pair is heterozygous.
- **Each gene pair in the F_2:** 3 dominant : 1 recessive, so counting dominant doses across all three genes gives the seven classes.
- **The F_2 class sizes:** 1 : 6 : 15 : 20 : 15 : 6 : 1 — darkest at one end, lightest at the other, most children in the middle classes.

Plot those seven numbers and you have drawn a bell. Add environmental effects — nutrition, sunlight — and the classes blur into a smooth spread. No Punnett square of 4 or 16 boxes can produce this: the shape **itself** is the evidence for polygenic inheritance.

Sex determination: four patterns

Pattern	Who decides what	The facts to quote
XX — XY (humans, most mammals)	The father's sperm decides — males are the **heterogametic** sex.	Females are XX, males are XY. Eggs all carry X; sperm carry X or Y. An X-bearing sperm gives a girl, a Y-bearing sperm a boy. The Y carries the SRY gene, the master switch for male development.
ZZ — ZW (birds)	The **mother's egg** decides — females are the heterogametic sex.	Males are ZZ, females are ZW. The pattern is the mirror image of humans — a standard contrast in questions.
XO type (grasshopper, some insects)	Sperm carrying X or not carrying X decides.	Females are XX; males have only one X (written XO) — the male has an odd chromosome number.
Haplo-diploid (honeybee)	Whether the egg is fertilised at all decides.	Males (drones) develop from **unfertilised** eggs by parthenogenesis and carry half the set — 16 chromosomes. Females (workers and queens) come from fertilised eggs and carry 32.

- **Notice the variety:** sex is not always a chromosome lottery — in honeybees it is a fertilisation lottery. Quote the numbers: 16 for the male bee, 32 for the female.
- **Why questions like birds:** students auto-write "male decides" from the human case; in birds it is the female. Slow down at the word **bird**.

WORKED EXAMPLE — DOES THE FATHER REALLY DECIDE THE SEX?

A couple has four children — three girls and a boy. Was the mother "responsible", as old family talk claims?

- **The mother's eggs:** every egg carries an X — she has no other sex chromosome to give.
- **The father's sperm:** half carry X, half carry Y.
- **Each child:** an X-sperm gives a girl, a Y-sperm gives a boy — an independent 1 : 1 draw every time.

So the father's sperm does decide each child's sex, and the 3 : 1 split in four children is ordinary chance — four draws of a fair coin. The honeybee goes one better and lets **fertilisation itself** decide, as the table above shows.

Pedigree analysis: the family tree of a trait

- **Definition:** a **pedigree** is a chart of a family's generations showing who shows the trait and who merely carries it.
- **The symbols:** a **square** is male, a **circle** is female; **shaded** shapes are affected individuals; a horizontal line joins parents, vertical lines drop to children; each generation sits on one row (Roman-numbered).
- **What it is used for:** tracing whether a trait is dominant or recessive, autosomal or X-linked — and predicting risks for the next child.
- **A quick tell:** an X-linked recessive trait strikes males far more often, because a single affected X is enough in an XY male.

Genetic Disorders: Mendelian and Chromosomal

A change in one letter of DNA, or one extra chromosome in a cell division, can rewrite a person's health. Genetic disorders fall into two big shelves — **Mendelian disorders**, caused by faulty single genes, and **chromosomal disorders**, caused by missing, extra or rearranged whole chromosomes. The syllabus lists the shelf and every book on it.

Mendelian disorders: one faulty gene

Disorder	Inheritance	What goes wrong	One feature to quote
Haemophilia	X-linked recessive	A clotting protein is missing, so blood clots very slowly or not at all after injury.	Affected males pass the gene to all daughters as **carriers** — Queen Victoria's family made it famous.
Sickle-cell anaemia	Autosomal; Hb^S allele recessive to Hb^A	A point mutation swaps one DNA letter: "GAG" → "GUG" in the messenger, so glutamic acid is replaced by	Red cells sickle and clog vessels; carriers (Hb^AHb^S) survive malaria better.

Disorder	Inheritance	What goes wrong	One feature to quote
		valine at position 6 of beta-globin.	
Thalassemia	Autosomal recessive	Faulty production of the alpha (chromosome 16) or beta (chromosome 11) globin chains of haemoglobin.	Severe anaemia needing transfusions — and it is **not** infectious, unlike malaria.
Phenylketonuria (PKU)	Autosomal recessive	An enzyme for breaking down phenylalanine is missing, so the amino acid piles up in blood and brain.	Managed by a phenylalanine-restricted diet from infancy.
Colour blindness	X-linked recessive	Defective red or green cone photoreceptor genes on the X chromosome.	Red-green confusion; far more common in males.

- **Sickle-cell's twist:** the Hb^SHb^S disease is harmful, yet the carrier (Hb^AHb^S) gains **malaria resistance** — a heterozygote advantage that keeps the allele in the population where malaria is common.
- **The X-linked pattern:** haemophilia and colour blindness are **criss-cross** traits — an affected male passes the allele to daughters (carriers), who pass it to grandsons.

Reading a pedigree in three lines

What you see in the tree	Likely inheritance	The one-line check
Trait in every generation; an affected parent has affected children.	Dominant (often autosomal)	No skipped generations; two unaffected parents have unaffected children.
Trait skips generations; unaffected parents can have affected children.	Recessive	Carriers exist; consanguineous marriages raise the risk.
Mostly males affected; trait passes mother → son, never father → son.	X-linked recessive	Fathers give their X only to daughters — the criss-cross pattern.

Chromosomal disorders: whole chromosomes out of count

- **Aneuploidy:** one chromosome is gained or lost, mostly by **nondisjunction** (a chromosome pair fails to separate in meiosis). Counts read 45, 47…
- **Polyploidy:** entire extra sets of chromosomes (3n, 4n) — nearly always lethal in humans, common in plants.

- **Down's syndrome: trisomy of chromosome 21** — 47 chromosomes. Features in one line: short stature, rounded face, small folds at the eye corners, a single palm crease, some intellectual disability.
- **Klinefelter's syndrome:** male with one extra X — 44 + XXY (47). Tall, sterile, with some feminine secondary characters (gynaecomastia).
- **Turner's syndrome:** female with a missing X — 44 + XO (45). Short, sterile, with underdeveloped secondary sex characters.
- **Read the numbers like a fingerprint:** 47 with XXY = Klinefelter; 45 with XO = Turner; 47 with an extra 21 = Down's. Questions swap these freely.

WORKED EXAMPLE — COLOUR BLINDNESS IN THE NEXT GENERATION

A mother is a carrier (X^CX^c) and the father is normal (X^CY). C is normal vision, c is colour blindness (X-linked recessive). What do the children face?

- **Mother's gametes:** X^C or X^c. **Father's gametes:** X^C or Y.
- **Daughters:** X^CX^C (normal) or X^CX^c (carrier) — none colour blind.
- **Sons:** X^CY (normal) or X^cY (colour blind) — a 50% chance each.

So 50% of the **sons** are expected to be colour blind and no daughters are — the classic X-linked recessive signature. A colour-blind father (X^cY) with a normal mother (X^CX^C) would pass X^c to **all** daughters — every one a carrier, and no son affected (sons got his Y).

Which disorder am I? Five clues

Clue in the question	Disorder	Lock the fact
"Boys bleed easily; the mother's father was also a bleeder."	Haemophilia	X-linked recessive — the criss-cross trail through carriers.
"One DNA letter changed; valine sits where glutamic acid should be."	Sickle-cell anaemia	"GAG" → "GUG" at position 6 of beta-globin.
"Three copies of chromosome 21; 47 chromosomes in all."	Down's syndrome	Trisomy 21 — usually from nondisjunction in meiosis.
"A tall, sterile male; 47 chromosomes with an extra X."	Klinefelter's syndrome	44 + XXY — the extra X shows in his build.
"A short, sterile female; only 45 chromosomes."	Turner's syndrome	44 + XO — one X chromosome short.

Key Facts & Terms Cheatsheet

KEY FACTS & TERMS CHEATSHEET — PRINCIPLES OF INHERITANCE AND VARIATION

Term / fact	What it means	Example / number
Law of dominance	In a contrasting pair of factors, one covers the other in the F_1.	$TT \times tt \rightarrow$ all tall F_1
Law of segregation	Factor pairs separate at gamete formation; each gamete carries one.	Tt plant $\rightarrow$ T and t gametes, 1 : 1
Law of independent assortment	One factor pair sorts into gametes independently of another pair.	Dihybrid $F_2 = 9 : 3 : 3 : 1$
Monohybrid F_2 ratios	Genotypes run 1 : 2 : 1; phenotypes run 3 : 1 (complete dominance).	$Tt \times Tt \rightarrow 1\ TT : 2\ Tt : 1\ tt$
Test cross	$F_1 \times$ recessive parent; a 1 : 1 result proves heterozygosity.	$Tt \times tt \rightarrow$ 1 tall : 1 dwarf
Back cross	F_1 crossed back to its own parent or a parent-like individual.	Breeders rebuild a parent's background
Incomplete dominance	F_1 is a blend; phenotypic ratio = genotypic ratio in the F_2.	Snapdragon red $\times$ white $\rightarrow$ pink, F_2 1 : 2 : 1
Co-dominance	Both alleles express together — nothing blends.	$I^A I^B \rightarrow$ blood group AB
ABO system	One gene, three alleles in the population; any person carries two.	6 genotypes $\rightarrow$ 4 phenotypes (A, B, AB, O)
Chromosomal theory	Chromosomes are the physical carriers of Mendel's factors.	Sutton and Boveri, 1902
Linkage	Genes on the same chromosome travel together in inheritance.	Tightly linked $\rightarrow$ fewer recombinants
Recombination	Crossing over in prophase I separates linked genes.	Frequency rises with gene distance
Morgan's white eye	First gene pinned to a specific chromosome — the X.	Drosophila white-eyed male
Polygenic inheritance	Many genes plus environment build one continuous trait.	Skin colour, height, wheat kernel colour
Sex determination	Humans XY (male decides); birds ZW (female decides, males ZZ); grasshopper XO; honeybee haplodiploid.	Male bee 16 chromosomes; female 32

Term / fact	What it means	Example / number
Sickle-cell mutation	One DNA letter swap changes the codon and the amino acid.	"GAG" → "GUG"; glutamic acid → valine, position 6
Down's syndrome	Trisomy of chromosome 21.	47 chromosomes (44 + 21 + 21 + 21)
Klinefelter's syndrome	Male with an extra X chromosome.	44 + XXY = 47, sterile male
Turner's syndrome	Female with one X missing.	44 + XO = 45, sterile female

Carry these numbers: 1856–1863 (the pea years), 1866 (published), 1900 (rediscovered), 1902 (chromosomal theory), 14,349 (seed-shape plants), 3 : 1 · 1 : 2 : 1 · 1 : 1 · 9 : 3 : 3 : 1 · 1 : 1 : 1 : 1 (the five ratios), 16 and 32 (honeybee), 45 · 47 (chromosome counts).

Common Errors and Special Cases

COMMON ERRORS & SPECIAL CASES

The mistake students make	What is right	One-line why / trap
Treating incomplete dominance and co-dominance as the same thing	Incomplete = blend (pink snapdragon); co-dominance = both show (AB blood).	A blended new shade is not the same as two products displayed side by side.
Quoting 3 : 1 as the snapdragon F_2 phenotypic ratio	1 red : 2 pink : 1 white — the phenotypic ratio is 1 : 2 : 1 here.	The genotype and phenotype ratios **coincide** in incomplete dominance — that coincidence is the trap.
Expecting an AB child from an AB parent × O parent	Only A (I^Ai) and B (I^Bi) children are possible.	The AB parent always passes I^A or I^B; the O parent has only i to give.
Calling group O a "missing" gene	O = ii — a real allele that simply makes no surface sugar.	i is recessive, not absent; two O parents make only O children.

The mistake students make	What is right	One-line why / trap
Expecting clean 3 : 1 ratios from polygenic traits	Many additive genes give a bell-shaped F_2 range.	Each gene pair segregates on its own; the products pile up instead of sorting into bins.
Swapping Turner's and Klinefelter's syndrome	Turner = XO, female, 45; Klinefelter = XXY, male, 47.	Both are aneuploid sex-chromosome counts — read the sex chromosome letters before answering.
Forgetting the sickle-cell carrier's advantage, or calling thalassemia X-linked	Carrier $Hb^A Hb^S$ resists malaria; thalassemia is autosomal recessive.	Heterozygote advantage is real; thalassemia is on autosomes 16 and 11, not the X.

The Whole Chapter on One Page



Figure 2: Whole chapter on one page — the six idea-clusters of Unit VII and the four traps that catch most students.

Final Drill: Complete the Ratio

Before the chapter map, run this 60-second self-check. Cover the middle column and fill each blank from memory — every line is a genuine question shape.

Cross or fact	Fill in	Check line
Monohybrid F_2 of $Tt \times Tt$: phenotypes?	3 : 1	Three tall looks, one dwarf — tt is the only dwarf.
Monohybrid F_2: genotypes?	1 : 2 : 1	One pure tall, two hybrids, one pure dwarf.
Snapdragon F_2 from two pink plants: phenotypes?	1 : 2 : 1	Red, pink, white — the blend makes genotypes visible.
Dihybrid F_2 of $RrYy \times RrYy$: phenotypes?	9 : 3 : 3 : 1	16 boxes: 9 + 3 + 3 + 1.
Dihybrid test cross of the F_1?	1 : 1 : 1 : 1	Four equal gamete types meet only ry.
Blood-group cross $I^Ai \times I^Bi$: children?	A, B, AB, O	All four groups — both parents carry the silent i.
Sickle-cell messenger change?	"GAG" → "GUG"	Glutamic acid is replaced by valine at position 6.

Five Question Shapes, Five Attack Plans

Genetics questions repeat their shapes — read the shape first, then reach for the matching tool.

- **Shape 1 — "The offspring ratio is given, find the parent."** A 1 : 1 split screams **test cross** (one parent heterozygous); a clean 3 : 1 points to two heterozygotes; all-tall offspring from a tall × dwarf cross point to a pure TT parent. Write the gametes before you write the answer.
- **Shape 2 — "Blood group puzzle."** Write the six ABO genotypes from memory first, then convert each stated group into its possible genotype pairs. Watch for the hidden i: groups A and B each carry two possible genotypes, and an O child from A × B parents is legal, while AB × O can never give O.
- **Shape 3 — "Count the dihybrid boxes."** Convert the total to sixteenths: round-yellow is 9/16, round-green 3/16, wrinkled-yellow 3/16, wrinkled-green 1/16. For 160 seeds that is 90, 30, 30 and 10 — one fraction line, no full grid needed.
- **Shape 4 — "Sex-linked family tree."** Colour blindness and haemophilia ride the X: a son's X comes only from his mother, so **her** carrier status decides the sons. Draw X's and Y's before touching the options.
- **Shape 5 — "Match the karyotype."** Count first (45, 47), then read the letters: XO = Turner (female), XXY = Klinefelter (male), +21 = Down's. The chromosome **count** and the **letters** together pin the answer.

Practice Questions

Q1. A father has blood group AB and the mother has blood group O. The blood groups possible in their children are:

(A) A or B only
(B) AB or O only
(C) All four groups
(D) Only AB

Q2. A tall pea plant is crossed with a dwarf plant, and half the offspring are tall while half are dwarf. The tall parent's genotype must have been:

(A) Tt
(B) TT
(C) tt
(D) Either TT or Tt

Q3. In an F_2 generation of 160 seeds from a RrYy $\times$ RrYy cross, the expected number of round-green seeds is:

(A) 10
(B) 20
(C) 30
(D) 90

Q4. In snapdragon, two pink-flowered plants (Rr) are crossed. The phenotypic ratio expected in the F_2 is:

(A) 3 : 1
(B) 9 : 3 : 3 : 1
(C) 1 : 1
(D) 1 : 2 : 1

Q5. A mother is a carrier of colour blindness (X^CX^c) and the father has normal vision (X^CY). What fraction of their **sons** is expected to be colour blind?

(A) 0%
(B) 25%
(C) 50%
(D) 100%

Q6. In the honeybee, the total number of chromosomes in a male drone is:

(A) 16
(B) 32
(C) 23
(D) 46

Q7. In sickle-cell anaemia, the point mutation changes the mRNA codon "GAG" into:

(A) "GUA"
(B) "GGG"
(C) "UAG"
(D) "GUG"

Q8. The chromosome complement of a human male with Klinefelter's syndrome is:

(A) 44 + XO
(B) 44 + XXX
(C) 44 + XXY
(D) 44 + YY

OurMocks

Q9. Two genes sit very close together on the same chromosome. A test cross involving them will produce offspring that are:

(A) Always in a 1 : 1 : 1 : 1 ratio
(B) Mostly parental combinations
(C) In a 9 : 3 : 3 : 1 ratio
(D) All recombinant types

Q10. Down's syndrome arises due to:

(A) Loss of one chromosome 21
(B) An extra copy of chromosome 21
(C) An extra X chromosome in a male
(D) A missing X chromosome in a female

Solutions

Sol. Q1. Write the cross: $I^{A}I^{B}$ (father) $\times$ ii (mother). The father's gametes carry I^{A} or I^{B}; the mother's carry only i. Children: $I^{A}i$ (group A) and $I^{B}i$ (group B), in a 1 : 1 ratio. An AB child needs both alleles from one parent — impossible; an O child needs two i — impossible from the father.

Answer: (A)

Sol. Q2. Half tall and half dwarf is a 1 : 1 ratio — the signature of a test cross. Dwarf offspring are tt, so both parents gave a t; the tall offspring are Tt. The tall parent therefore made both T and t gametes, so it was Tt. A TT parent would have given 100% tall offspring.

Answer: (B)

Sol. Q3. Round-green means at least one R with yy — the R_yy class, which is 3 boxes of the 16-box grid. Expected count = $3/16 \times 160 =$ **30 seeds**. (Round-yellow would have been 90, wrinkled-yellow 30, wrinkled-green 10.)

Answer: (C)

Sol. Q4. Pink $\times$ pink is $Rr \times Rr$: gametes R and r from each side give $1RR$ (red) : $2Rr$ (pink) : $1rr$ (white). In incomplete dominance every genotype wears its own phenotype, so the phenotypic ratio equals the genotypic ratio — 1 : 2 : 1, not 3 : 1.

Answer: (D)

Sol. Q5. Sons receive the Y from the father, so their vision depends only on the mother's X: X^{C} gives a normal son, X^{c} gives a colour-blind son ($X^{c}Y$). The mother makes both in equal numbers, so 50% of sons are colour blind. (Daughters would be normal or carriers — none colour blind.)

Answer: (C)

Sol. Q6. Honeybee males develop from unfertilised eggs by parthenogenesis, so a drone carries a single (haploid) set — 16 chromosomes. Females come from fertilised eggs and carry the full 32.

Answer: (A)

Sol. Q7. The sickle-cell point mutation replaces one DNA letter, converting the mRNA codon "GAG" to "GUG". "GUG" codes for valine instead of glutamic acid, which is inserted at position 6 of the beta-globin chain — the change that makes red cells sickle.

Answer: (D)

Sol. Q8. Klinefelter's is an aneuploid **male** condition with one extra X: 44 autosomes + XXY = 47 chromosomes. 44 + XO (45) is Turner's syndrome, a female condition — the pair students most often swap.

Answer: (C)

Sol. Q9. Tightly linked genes are rarely separated by crossing over, so the F_1 passes on the parental gene combinations far more often than new ones — the test cross yields **mostly parental types**, not the 1 : 1 : 1 : 1 of independently assorting genes.

Answer: (B)

Sol. Q10. Down's syndrome is trisomy of chromosome 21 — three copies instead of two, giving 47 chromosomes in total. It arises most often by nondisjunction of chromosome 21 during meiosis. (Extra X in a male = Klinefelter; missing X in a female = Turner.)

Answer: (B)

Ratios side by side — the last look

Cross situation	Ratio	Where it comes from
$Tt \times Tt$ — phenotypes	3 : 1	F_2 of the monohybrid under complete dominance.
$Tt \times Tt$ — genotypes	1 : 2 : 1	The four Punnett boxes, counted.
$Tt \times tt$ — the test cross	1 : 1	The recessive partner masks nothing.
Two pink snapdragons	1 : 2 : 1	Incomplete dominance lets genotypes show.
$RrYy \times RrYy$	9 : 3 : 3 : 1	Sixteen boxes, two characters at once.
$RrYy \times rryy$ — dihybrid test cross	1 : 1 : 1 : 1	Four gamete types, each in equal numbers.

© OurMocks · www.ourmocks.com · CUET UG Biology · Principles of Inheritance and Variation · 2027 cycle